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MILITARY SPECIFICATION

GRENADE, RIFLE, SMOKE, M22A2; CHEMICAL LOADING ASSEMBLY

This specification has been approved by the Department of Defense and is mandatory for use by the Departments of the Army, the Navy, and the Air Force.

1. SCOPE

1.1 This specification covers the loading and assembly (without fuzing components) of the M22A2 impact type rifle grenade with green, red, or yellow smoke mixture.

2. APPLICABLE DOCUMENTS

2.1 The following specifications, standards, drawings, and publication of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

FEDERAL

CCC-B-801 — Bunting; Wool.
CCC-S-271 — Sheeting; Cotton, Bleached, Wide.
CCC-S-291 — Sheeting; Cotton, Unbleached, Wide.
LLL-F-291 — Fiberboard, Corrugated, Single Face (Flexible).

MILITARY

JAN-P-106 — Packaging and Packing for Overseas Shipment — Boxes; Wood, Nailed.
JAN-P-127 — Packaging and Packing for Overseas Shipment — Tape, Adhesive, Pressure-Sensitive, Water Resistant.

STANDARDS

MILITARY

MIL-STD-105 — Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 — Marking for Shipment and Storage.
MIL-STD-508 — Formula for Starter Mixture—VI (with Binder).
MIL-STD-509 — Formula for Starter Mixture — V (with Binder).
MIL-STD-513 — Formula for Impregnating Mixture — I.
MIL-STD-526 — Formula for Green Smoke — V.
MIL-STD-529 — Formula for Red Smoke — V.
MIL-STD-532 — Formula for Yellow Smoke — VII.

FSC 1330

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DRAWINGS

ORDNANCE CORPS

75–14–448 — Grenade, Rifle, Smoke (Colored), M22A2 — Loading Assembly and Marking Diagram.

82–0–117 — Grenade, Rifle, Smoke (Colored), M22A2 — Assembly.

PUBLICATIONS

CHEMICAL CORPS DIRECTIVE

488 — Filling and Assembling Grenade, Rifle, Smoke Colored, M22A1.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 **Materials and components.** All materials, components, and subassemblies used in the loading assembly shall conform to the specifications, standards, and drawings forming a part of this specification.

3.2 **Filling.** The grenade shall be filled with a colored smoke charge (see 6.5), consolidated in one or more equal increments under a dead load pressure of not less than 3,000 pounds to a level 0 to ⅛ inch below the bottom of the body flange. A tapered hole, ½ inch in diameter at the nose of the grenade and ¼ inch at the base, shall be formed during the consolidating operation.

3.2.1 *Green smoke.* When green smoke is specified (see 6.2), the grenade shall be filled with a dry uniform mixture conforming to Standard MIL–STD–526.

3.2.2 *Red smoke.* When red smoke is specified (see 6.2), the grenade shall be filled with a dry uniform mixture conforming to Standard MIL–STD–529.

3.2.3 *Yellow smoke.* When yellow smoke is specified (see 6.2), the grenade shall be filled with a dry uniform mixture conforming to Standard MIL–STD–532.

3.3 **Starter mixture.** The top surface of the smoke mixture and the sides of the center tapered hole shall be coated with approximately 5 grams of a uniform mixture conforming to Standard MIL–STD–509 and dried thoroughly for a minimum of 15 minutes at not less than 75°F. Approximately 6 grams of a uniform mixture conforming to Standard MIL–STD–508 may be used as an alternate starter mixture.

3.4 **Starter cup cover.** The starter cup cover shall be prepared from bunting conforming to Specification CCC–B–801; or from sheeting which conforms to class B of Specification CCC–S–271 or of Specification CCC–S–291 except that the total sizing, finishing, and other nonfibrous materials shall not exceed 1.0 percent. The bunting or sheeting shall be thoroughly impregnated with a uniform mixture conforming to Standard MIL–STD–513, dried at room temperature for at least 30 minutes and then dried to constant weight in an oven at 100° to 150°F. prior to being cut into disks. A 0.0625-inch diameter hole shall be punched in the center of the impregnated disk to facilitate ignition.

3.5 **Assembly.** The body flange shall be cleaned prior to assembling the starter cup cover in the grenade. Closing, coating, marking, and assembly shall be in accordance with Drawing 75–14–448. Care shall be taken during the crimping operation to insure that the body union is seated squarely on the body flange before and after crimping. Water resistant adhesive tape, conforming to type I, grade A of Specification JAN–P–127, shall be used to seal the threaded opening and the holes in the body union.

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3.6 **Functioning.** When tested as specified in 4.4.1, the grenade shall ignite within 15 seconds after ground impact and emit a cloud of smoke of the color specified in the procurement document for not less than 30 seconds nor more than 2 minutes. The smoke shall be distinguishable as to color, observed against a background of contrasting color at a distance of 3,000 feet on a clear day. Flaming shall not exceed 20 percent of the total burning time.

3.7 **Pilot lot.** A pilot lot shall be produced in accordance with 4.1.1.

3.8 **Workmanship.** The grenade shall not be ruptured, deformed, or contaminated.

4. QUALITY ASSURANCE PROVISIONS

4.1 Lot.

4.1.1 *Pilot.* Prior to the start of regular production, and following any change in regular production methods, materials, or equipment, a pilot lot of 500 grenades shall be produced using the same methods, materials, and equipment as will be employed during regular production.

4.1.2 *Regular production.* A lot shall consist of the grenades produced by one manufacturer in not more than one work shift from the same materials with no change in formulation of starter and smoke mix, and under essentially the same manufacturing conditions. The inspection lot for nondestructive inspection, within the above limitations, shall be designated by the inspector. When specified in the contract or order, groups for shipment shall be formed from each production lot. The number of units in the group and the marking shall be as specified (see 6.2).

4.2 Sampling.

4.2.1 *For nondestructive inspection.*

4.2.1.1 *Pilot lot.* Each grenade in the pilot lot shall be inspected for conformance to the applicable drawings and specification requirements.

4.2.1.2 *Regular production lot.* Sampling shall be conducted in accordance with the classification of defects (4.3.3) and Standard MIL–STD–105.

4.2.2 *For functioning test.*

4.2.2.1 *Pilot lot.* Fifty representative grenades shall be taken from each pilot lot for testing as specified in 4.4.1.

4.2.2.2 *Regular production lot.* Sixty representative grenades, 30 for test and 30 for retest, shall be taken from each production lot for testing as specified in 4.4.1.

4.3 Inspection.

4.3.1 *Materials and components.* The inspector shall ascertain that all materials, components, and subassemblies of the grenade, covered by separate drawings, specifications, or standards, are in accordance with the applicable drawings, specifications, or standards prior to final acceptance of the loaded assembly.

4.3.2 *Packing and marking.* The inspector shall ascertain that the packing and marking of the grenade conform to this specification.

4.3.3 *Classification of defects.* Inspection of the grenade shall be conducted as follows:

4.3.3.1 *Grenade, rifle, smoke (colored), M22A2 loading assembly.* (Dwg 75–14–448)

Categories and defects

Major — *AQL 1.0 percent defective*

1. Component missing or incorrectly assembled
2. Filling inadequate
3. Union improperly crimped to body or cement missing
4. Marking incorrect
5. Starter charge inadequate
6. Starter cup cover impregnation missing or inadequate
7. Damage
8. Starter charge hole incorrectly shaped

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Minor *AQL 2.5 percent defective*

51. Marking missing or illegible
52. Coating inadequate or incorrect

Incidental *AQL 10 percent defective*

101. Contamination

4.4 Functioning test. The grenade samples shall be assembled in accordance with Drawing 82–0–117. Each grenade shall be fired from a rifle at an angle of 45° from the horizontal in accordance with standard procedure for a munition of this type. The following information shall be determined and recorded:

(a) Lag time (between impact and smoke emission).

(b) Time smoke is emitted under pressure.

(c) Total flaming time.

(d) Color of smoke from a distance of 3,000 feet.

Grenades that fail due to any of the following causes shall not be considered in the functioning test but shall be replaced by representative grenades from the same lot:

(a) Faulty primer or tail assembly.

(b) Landing in water or soft mud.

(c) Splitting of the grenade on impact with the ground.

4.5 Rejection.

4.5.1 *Pilot lot.* If any grenade fails to meet the requirements of the specification, the pilot lot represented shall be rejected. The cause of failure shall be ascertained and corrective action instituted prior to initiation of a new pilot lot. Upon rejection of a pilot lot, a new pilot lot shall be produced and subjected to the same sampling and inspection as the original pilot lot. Regular production, pending approval of the pilot lot, will be at the risk of the manufacturer.

4.5.2 *Regular production lot.* The regular production lot, when tested as specified in 4.4.1, shall be disposed of in accordance with table I.

TABLE I. *Rejection criteria*

Attribute	Test criteria			Retest criteria		
	Sample	Failures (test)	Action	Sample	Failures (test plus retest)	Action
Incorrect color*	30	0 1 or more	Accept Reject	30	0 1 or more	Accept Reject
Dud**	30	0 1 or 2 3 or more	Accept Retest Reject	30	1 or 2 3 or more	Accept Reject
Off-color*** Smoke emission time *Flamers**** or* lag time*****	30	0, 1, or 2 3 or 4 5 or more	Accept Retest Reject	30	3 or 4 5 or more	Accept Reject

* Color other than that marked on the grenade.

** Grenade that fails to ignite.

*** Off-color grenade is defined as a grenade whose smoke is of the correct color but not of sufficient volume, rendering it nondistinguishable as to color when viewed against a background of contrasting color from a distance of 3,000 feet on a clear day.

**** Grenades which flame at one end shall not be considered flamers if an effective smoke of the required color issues from the other end.

***** Time between impact and smoke emission.

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5. PREPARATION FOR DELIVERY

5.1 Packing.

5.1.1 *Level B.* Filled and closed grenade bodies shall be packed in boxes conforming to Specification JAN–P–106, the inside dimensions of which shall be approximately 13 by 13 by 12 inches. The bodies shall be packed in the boxes in two layers of 36 bodies each. Corrugated flexible fiberboard conforming to Specification LLL–F–291 shall be used to line the box and to separate and cover each layer. The bodies in each layer shall be individually separated and packed in half-slotted, interlocking, extended tip partitions.

5.1.2 *Level C.* When the filled and closed grenade bodies are to be transferred from the filling plant to a local Ordnance Corps assembly plant, they shall be packed in the same manner as the empty grenade bodies were received.

5.2 **Marking.** In addition to any special marking required by the contract or order, all shipping containers shall be marked in accordance with Standard MIL–STD–129 and figure 1.

6. NOTES

6.1 Intended use. The grenades covered by this specification are intended for use in signaling.

6.2 Ordering data. Procurement documents should specify:

(a) The title, number, and date of this specification.

(b) The color of smoke required.

(c) The shipment group size and marking (when required).

(d) The level of packing required.

6.3 Precautions. Because of the hazardous nature of filling munitions with chemical agents, filling installations are advised to take all necessary precautions to protect personnel from injury while filling grenades in accordance with this specification.

6.4 Directive. Directive No. 488 contains a suggested method of loading the grenade.

6.5 Filling aid. Grenades filled with approximately 0.4 pound of colored smoke mix have been found to comply with 3.2.

Patent notice. When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

Custodians:
Army—Chemical Corps
Navy—Bureau of Ordnance

Other interest:
Army—O

Preparing activity:
Army—Chemical Corps

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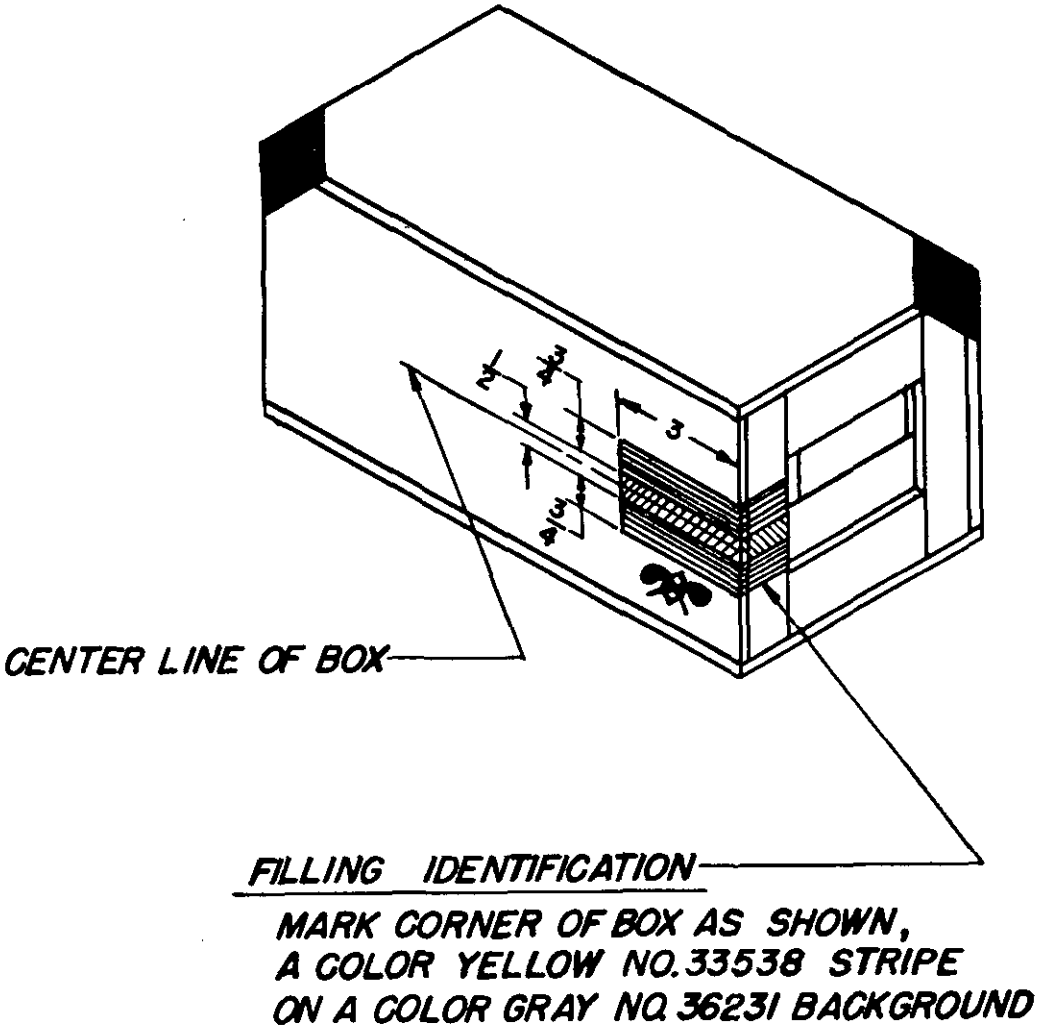


FIGURE 1. *Filling identification.*

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